



RAMCO INSTITUTE OF TECHNOLOGY

APPROVED BY **AICTE** AND AFFILIATED TO **ANNA UNIVERSITY**,
ACCREDITED BY **NAAC**, AN ISO 9001:2015 CERTIFIED)



DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

ABOUT THE DEPARTMENT



The Department of Electronics and Communication Engineering was established in the year 2013 with a vision to develop competent professionals to meet the global challenges in modern engineering society. At present it offers a 4-year B.E. Electronics and Communication Engineering programme with a sanctioned intake of 120 students. The Laboratories are equipped with latest technology in the fields of Electronic Devices and Circuits, Digital Electronics, Linear Integrated Circuits, Microprocessors and Microcontrollers, Digital Signal Processing, Communication System, VLSI, Embedded Systems, Optical & Microwave, Embedded systems that offer opportunities on a wide range of hardware and advanced software packages. In addition to this, the Department has equipped with Research Laboratories such as NI-LabVIEW Research Laboratory, Tessolve Semiconductor Laboratory and Technical Incubation Centre to enrich the knowledge of students and faculty. The Department has a team of committed faculty members with sound knowledge in the areas of Applied Electronics, Communication systems, VLSI, Embedded Systems, Computer Communication and Networking. The Department encompasses professional associations such as Students Technical Association Electronica, IETE, ISTE chapters which provide students the platform to learn professional and soft skills. The Department delights to release the Technical Magazine - "ELECOM MAGAZINSIO" for the academic year 2021-2022 Odd Semester





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VISION

To evolve as an Institute of international repute in offering high-quality technical education, Research and extension programmes in order to create knowledgeable, professionally competent and skilled Engineers and Technologists capable of working in multi-disciplinary environment to cater to the societal needs.

MISSION

- To accomplish its unique vision, the Institute has a far-reaching mission that aims:
- To offer higher education in Engineering and Technology with highest level of quality, professionalism and ethical standards
- To equip the students with up-to-date knowledge in cutting-edge technologies, wisdom, creativity and passion for innovation, and life-long learning skills
- To constantly motivate and involve the students and faculty members in the education process for continuously improving their performance to achieve excellence

QUALITY POLICY

RIT aims to impart Quality Education in Engineering and Technology through an effective teaching-learning process, up-gradation of facilities and human resources, collaborating with Industry for promoting training and placement, research and consultancy activities with a commitment to continual improvement of Quality Management System.



ELECTRONICS AND COMMUNICATION ENGINEERING

VISION

Develop futuristic technologist in Electronics and Communication Engineering to meet the global challenges and the societal needs of our nation.

MISSION

- Educate students to become enlightened engineers with the latest know-how to take on the challenges concerned with the society.
- Equip the students with the current trends and latest technologies in the field of Electronics & Communication Engineering to make them technically strong and ethically sound.
- Constantly motivate the students and faculty members to achieve excellence in Electronics & Communication Engineering and Research & Development activities.

PROGRAM EDUCATIONAL OBJECTIVES (PEOs) :

- Graduates will gain knowledge through continuous learning to enhance competency and career prospects.
- Graduates will be technically competent to design, analyze, develop and implement Electronic and Communication Systems.
- Graduates will apply the basic fundamental concepts in Electronic and related fields to solve general engineering problems related to societal needs.
- Graduates will be professionals with attributes, passion, positive attitude, ethics and moral values for a successful career.
- Graduates will Communicate effectively and interact responsibly with colleagues, employees, team members, clients and the society.

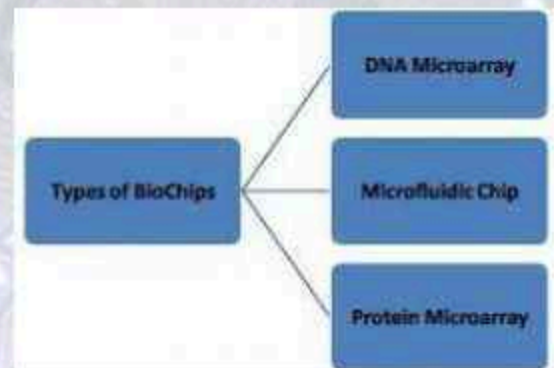
PROGRAM SPECIFIC OUTCOMES (PSOS) :

After successful completion of the degree, the students will be able to:

- Adapt to emerging trends in Information and Communication Technology by innovating new ideas to solve existing/novel problems.
- Excel in latest technologies of Embedded Systems.
- Design, analyze and develop electronic products in the area of VLSI Design and Communication Systems.
- Contribute as Entrepreneurs in building electronic products.

Bio-Chip Technology

A BIOCHIP IS A COLLECTION OF MINIATURIZED TEST SITES (MICROARRAYS) ARRANGED ON A SOLID SUBSTRATE THAT PERMITS MANY TESTS TO BE PERFORMED AT THE SAME TIME IN ORDER TO ACHIEVE HIGHER THROUGHPUT AND SPEED. TYPICALLY, A BIOCHIP'S SURFACE AREA IS NO LARGER THAN A FINGERNAIL. LIKE A COMPUTER CHIP THAT CAN PERFORM MILLIONS OF MATHEMATICAL OPERATIONS IN ONE SECOND, A BIOCHIP CAN PERFORM THOUSANDS OF BIOLOGICAL REACTIONS, SUCH AS DECODING GENES, IN A FEW SECONDS.



A BIOCHIP COMPRISES MAINLY THREE TYPES: DNA MICROARRAY, PROTEIN MICROARRAY, AND MICROFLUIDIC CHIP.

Applications of BIO-CHIP

THE APPLICATIONS OF BIOCHIP INCLUDE THE FOLLOWING. BY USING THIS CHIP WE CAN TRACE A PERSON OR ANIMAL ANYWHERE IN THE WORLD. THIS CHIP IS USED TO STORE AND UPDATE THE INFORMATION OF A PERSON LIKE MEDICAL FINANCIAL AND DEMOGRAPHICS. THESE CHIPS ARE EFFECTIVE IN RESTORING THE RECORDS OF MEDICAL, CASH, PASSPORT, ETC.

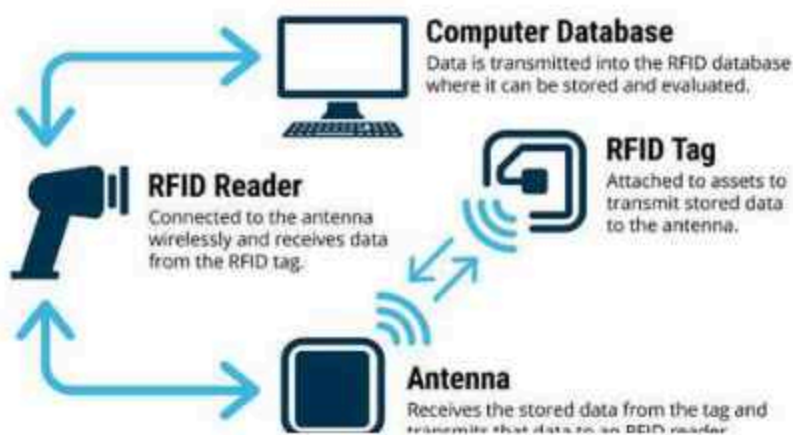
Harikiran S

III YEAR ECE - A



RADIO FREQUENCY IDENTIFICATION (RFID) REFERS TO A WIRELESS SYSTEM COMPRISED OF TWO COMPONENTS: TAGS AND READERS. THE READER IS A DEVICE THAT HAS ONE OR MORE ANTENNAS THAT EMIT RADIO WAVES AND RECEIVE SIGNALS BACK FROM THE RFID TAG. TAGS, WHICH **USE RADIO WAVES TO COMMUNICATE THEIR IDENTITY AND OTHER INFORMATION TO NEARBY READERS**, CAN BE PASSIVE OR ACTIVE. PASSIVE RFID TAGS ARE POWERED BY THE READER AND DO NOT HAVE A BATTERY. ACTIVE RFID TAGS ARE POWERED BY BATTERIES.

Basic RFID System



Uses

RFID SYSTEMS USE RADIO WAVES AT SEVERAL DIFFERENT FREQUENCIES TO TRANSFER DATA. IN HEALTH CARE AND HOSPITAL SETTINGS, RFID TECHNOLOGIES INCLUDE THE FOLLOWING APPLICATIONS:

- INVENTORY CONTROL
- EQUIPMENT TRACKING
- OUT-OF-BED DETECTION AND FALL DETECTION
- PERSONNEL TRACKING
- ENSURING THAT PATIENTS RECEIVE THE CORRECT MEDICATIONS AND MEDICAL DEVICES
- PREVENTING THE DISTRIBUTION OF COUNTERFEIT DRUGS AND MEDICAL DEVICES
- MONITORING PATIENTS
- PROVIDING DATA FOR ELECTRONIC MEDICAL RECORDS SYSTEMS

THE TWO MAIN PRIVACY CONCERNS REGARDING RFID ARE AS FOLLOWS:

- AS THE OWNER OF AN ITEM MAY NOT NECESSARILY BE AWARE OF THE PRESENCE OF AN RFID TAG AND THE TAG CAN BE READ AT A DISTANCE WITHOUT THE KNOWLEDGE OF THE INDIVIDUAL, SENSITIVE DATA MAY BE ACQUIRED WITHOUT CONSENT.
- IF A TAGGED ITEM IS PAID FOR BY CREDIT CARD OR IN CONJUNCTION WITH USE OF A LOYALTY CARD, THEN IT WOULD BE POSSIBLE TO INDIRECTLY DEDUCE THE IDENTITY OF THE PURCHASER BY READING THE GLOBALLY UNIQUE ID OF THAT ITEM CONTAINED IN THE RFID TAG. THIS IS A POSSIBILITY IF THE PERSON WATCHING ALSO HAD ACCESS TO THE LOYALTY CARD AND CREDIT CARD DATA, AND THE PERSON WITH THE EQUIPMENT KNOWS WHERE THE PURCHASER IS GOING TO BE.

Deepak Chakravarthi

III YEAR ECE - A

PHOTOGRAPHY



P.V.Surendar

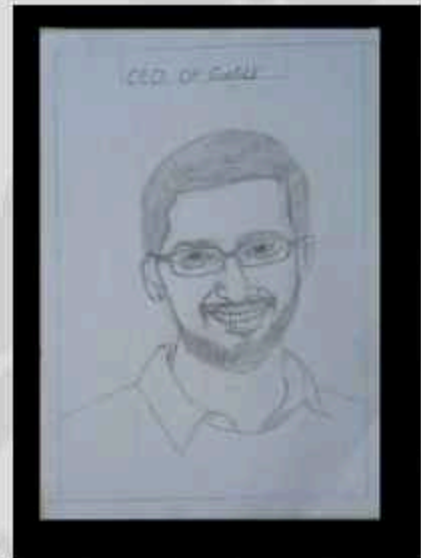
III YEAR ECE - B



Vijayarajamohan

III YEAR ECE - B

PENCIL SKETCH



Shriram
I YEAR ECE

PHOTOGRAPHY



B. Divyashri

II YEAR ECE



INFORMATION AND NETWORK SECURITY

DATA IS AN UNORGANIZED FACT THAT CAN BE PROCESSED INTO MEANINGFUL INFORMATION..WHEREAS INFORMATION IS A ORGANIZED FACT ,INTEREPTED AND MADE INTO MEANING THING.**HACKING IS AN UNAUTHORIZED ENTRY INTO A NETWORK OR COMPUTER TO STEAL INFORMATION , DATA OR FILES.**Types of Hackers are like White Hat ,Gray Hat and Black Hat hacker. Malware is derived from malicious software, which will slow down the system.

PHISHING IS THE FRADULENT ATTEMPT TO OBTAIN SENSITIVE INFORMATION, PASSWORDS, AND CREDIT CARD DETAILS.To set a strong or the secure password in that encryption plays a vital role in securing our data to keep it safe. It is an activity designed to protect the usability and integrity of your network and data. It includes both hardware and software. A fraudulent wireless access point to steal the information.It also leads to eavesdropping without the users knowledge



- **PUBLIC FREE WIFI**
- **FREE CHARGING STATION.**
- **ZOMATO APP**
- **CAMSCANER.**



It is a type of a malware .It is a horse shape symbol. It is a email viruses that duplicate themselves, steal information or harm computer system

PENCIL SKETCH

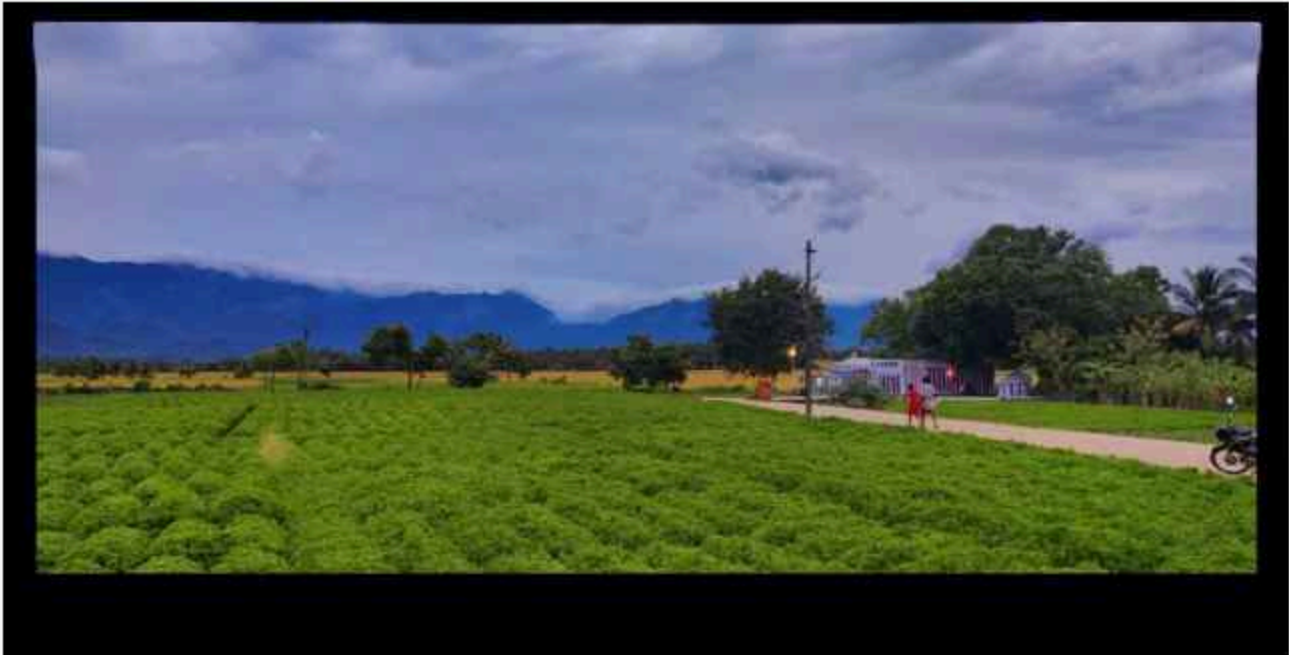


Uma Maheswari
I YEAR ECE

Srinivass
I YEAR ECE



PHOTOGRAPY



S. Aarthi

II YEAR ECE

SMART LIGHTING SYSTEM

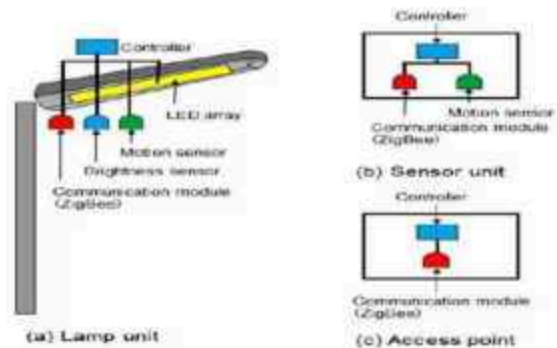
- SLS is the good way which enables to minimize and save light by allowing the householder to control remotely cooling and heating, lighting, and the control of appliances.
- The system serve to provide the right amount of light where and when it is needed



Day light dimming : Controls the artificial light depending on the available daylight .

Occupancy sensors : Automatically turns ON the lights when the room is occupied and OFF the light when the room is vacant .

Wireless lighting : Lamps are controlled by remote controls, wall switches or internet connected devices .



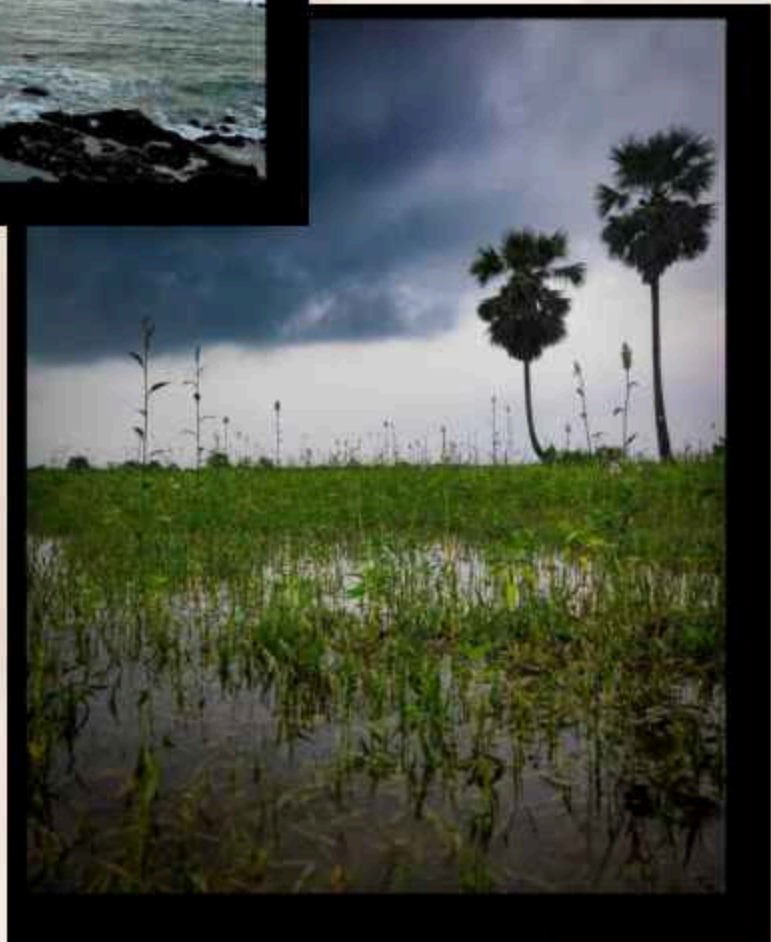
ADVANTAGES

- Reduction of energy and maintenance costs
- Increased public safety from improved lighting
- Safety traffic due to increased visibility of hazards
- Measurable environmental impact due to reduced energy consumption

S. Lingesh

III YEAR ECE - A

PHOTOGRAPHY



K. Aparna
II YEAR ECE

PENCIL SKETCH

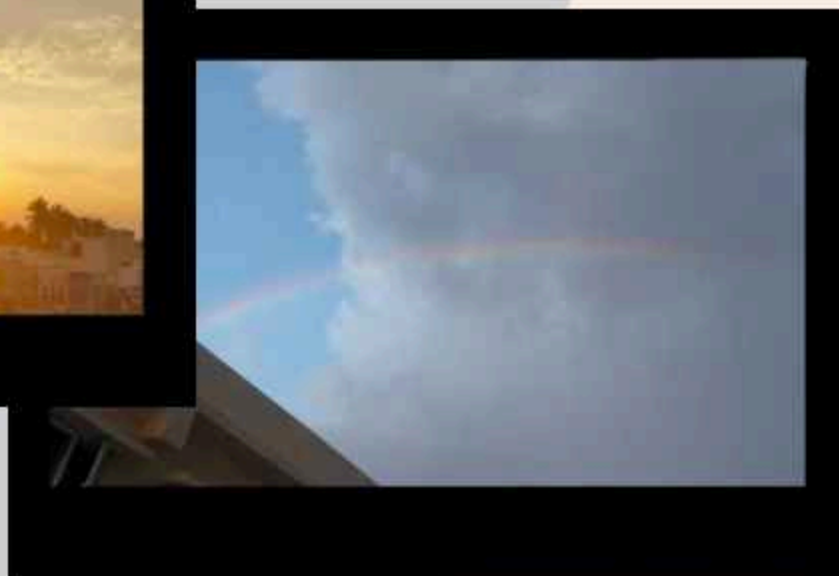


P. Madhumitha
I YEAR ECE

PHOTOGRAPHY



Azibamina
III YEAR ECE - A



PENCIL SKETCH



Sathya
I YEAR ECE

PHOTOGRAPHY



Vijayarammen
III YEAR ECE-A

QUOTES

“

I feel safe when you are with me;
You show me fun things to do;
you make my life much better;
The best father I know is you;

LOVE YOU DAD

I will always with you...



Now it is time to put yourself
Let go off all the bad times,
Switch off the negativity
Focus on your goals
Reach for the moon
Do whatever Pleases you,
You are the master of your own destiny...

Rajalakshmi R

III YEAR ECE-A

அண்ணன்

பெண்களின் மனம் ஆழமாம்..
அண்ணனின் மனதை
அளந்து பார்க்காதவர்கள்
சொல்லிக் கொள்ளுகிறார்கள்..!
அண்ணன் என்பவன் கடவுளால்
கொடுக்கப்பட்ட
அற்புதமான வரம்..
அந்த வரத்தை பெற்றவர்களுக்கு
அப்பாசத்தின் விலைமதிப்பு
தெரியாது.. அந்த வரம்
கிடைக்காமல் அந்த
பாசத்திற்காக ஏங்குபர்களுக்கு
மட்டும் தான் தெரியும்
அப்பாசத்திற்கு விலை மதிப்பே
இல்லை என்று..!
எப்போது சண்டை போட்டாலும்
என் பொய்யான அழகையில்
அனைத்தையும் விட்டுத் தரும்
நல்ல உறவு என் அண்ணன்..!

Muthumeena

II-ECE B

SARCO-SUICIDE CAPSULE



Some 1,300 people died by assisted suicide in Switzerland in 2020 using the services of the country's two largest assisted suicide organisations, Exit (no connection to Exit International) and Dignitas. The method currently in use is ingestion of liquid sodium pentobarbital.

After taking the drug, the person will fall asleep within two to five minutes before slipping into a deep coma, followed soon afterwards by death. Sarco offers a different approach for a peaceful death, without the need for controlled substances.

It's a 3-D printed capsule, activated from the inside by the person intending to die. The machine can be towed anywhere for the death. It can be in an idyllic outdoor setting or in the premises of an assisted suicide organisation, for example.

The person will get into the capsule and lie down. It's very comfortable. They will be asked a number of questions and when they have answered, they may press the button inside the capsule activating the mechanism in their own time.



The capsule is sitting on a piece of equipment that will flood the interior with nitrogen, rapidly reducing the oxygen level to 1 per cent from 21 per cent in about 30 seconds. The person will feel a little disoriented and may feel slightly euphoric before they lose consciousness.

Kezava anand

III YEAR ECE-A



P. Kalidas

III YEAR ECE-A

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